To verify that multiplication of integers is associative, let us check whether $(-5)\times\left[(-9)\times(-12)\right]$ and $\left[(-5)\times(-9)\right]\times(-12)$ are equal.

$(-5)\times\left[(-9)\times(-12)\right]=(-540)$ and $\left[(-5)\times(-9)\right]\times(-12) = (-540)$.

Hence, "associative" property is true for mutliplication of integers.

Therefore, for any three integers a, b, c ; $(a\times b)\times c = a\times(b\times c)$.

Just as zero added to a integer leaves it unchanged, integer 1 multiplied with any integer leaves the integer unaltered. For example, $57\times1=57$, and $1\times(-62)=-62$. We say that the integer 1 is the identity for multiplication of integers or "multiplicative identity".

Therefore, for any integer a; $a \times 1 = 1 \times a = a$.

Try these

1. Find the product and check for equality :
   (i) $18\times(-5)$ and $(-5) \times 18$
   (ii) $31 \times (-6)$ and $(-6) \times 31$
   (iii) 4×51 and 51×4
2. Prove the following :
   (i) $(-20) \times (13\times4) = \left[(-20)\times13\right]\times4$
   (ii) $\left[(-50)\times(-2)\right]\times(-3)=(-50)\times\left[(-2)\times(-3)\right]$
   (iii) $\left[(-4)\times(-3)\right]\times(-5)=(-4)\times\left[(-3)\times(-5)\right]$

Note

Consider an example, $(-7) \times (-6) \times (-5) \times (-4)$.

Let us try to do the above multiplication of integer,

$$\begin{aligned}(-7) \times (-6) \times (-5) \times (-4) &= [(-7) \times (-6)] \times [(-5) \times (-4)]\\ &= (+42) \times (+20)\\ &= +840\end{aligned}$$

From the above example, we see that the product of four negative integers is positive. What will happen if we multiply odd number of negative integers.

Let us consider another example, $(-7) \times (-3) \times (-2)$.

Multiplying the above integers, we get

$$\begin{aligned}(-7) \times (-3) \times (-2) &= [(-7) \times (-3)] \times (-2)\\ &= (+21) \times (-2)\\ &= -42\end{aligned}$$

From the above example, we see that the product of three negative integers is negative.

In general, if the negative integers are multiplied even number of times, the product is a positive integer, whereas negative integers are multiplied odd number of times, the product is a negative integer.

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